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- Loading Data

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The first step in your data science project is to load data into Pulse and begin your [data science loop](#).

Before starting the steps, ensure that the sample data file is downloaded to the `PULSE_SERVER_HOME/var/datasources/` path on the Pulse server where you are working. To get help with this, get in touch with your contact person at Feedzai.

To load a dataset into Pulse, execute the following actions:

1. In a browser, navigate to your data science environment. Log in by using the credentials provided, which by default are **pulse/pulse**.
2. Create a new application named *Fraud Training*.
3. In the **Data Science** tab, add a new [data source](#) named *RawTransactions*.

The URL depends on the operating system of your Pulse server and the path to the Pulse server home. For example: `file:///PULSE_SERVER_HOME/var/datasources/transactions.csv`.

4. Examine the dataset by scrolling through the different columns and understanding what the dataset contains.

1. Note that there are different ways this dataset represents missing values, such as using `-` in *AVSAddressNumber* or `\N` in *AuthSequenceNumber*.
2. Add both `-` and `\N` to the **Missing Values** list.
3. Proceed to the **Next** step.

5. After Pulse has inferred the data type for each field, in the **Data Source Characterization** step, check if all the fields have suitable [data types](#) and correct those that don't.

1. Change the type of some of the amount fields from *Categorical* to *Numeric*, such as *AdditionalAmount1* and several others, whenever you think it may make sense.

Be careful not to change the data type of true categoricals such as *CashAdvanceAmount* (does not contain numbers) or *AVSZipCode* (even though it has numbers, it can be argued there is not an order in zip codes).

2. Change the type of *ID* fields (such as *AccountID* or *RouterID*) to have a *String* type rather than *Numeric* or *Categorical*, since it was not correctly inferred.
3. Change the type of *RecurringBillingKey* and *InternetAuthorizationIndicator* to *Date Time*. Set the format of these fields to *Date Time*, with *dd/MM/yyyy* format.
4. Use the button to change the data type configuration of the *AdditionalAmount1* field that you set up earlier. It is currently an *Integer* field. Set it to be *Float*. This makes sense because *AdditionalAmount1* has decimal places and you don't want to lose information.

6. Scroll down to the bottom of the page and create your improved data source.

7. Select your data source from the **Data Sources** list.

8. Use the **Analyse** button to analyze the data source. Select *Fraud* as a breakdown field. This way you can analyze correlations between fields of the data set, in particular, the *Fraud* field, which is the domain goal in the scope of this application.

1. Use the **View Statistics** option to better understand fields in your dataset, including arithmetic statistics of the field and its value distribution. It is also possible to click over the histogram to access the statistics view.
2. View the statistics of the *TransactionAmount* field by using the button. Break the analysis down by *Fraud* to observe that the mean value of *TransactionAmount* is greater for fraudulent transactions.